ANURAG KUMAR SAHU

Korba, Chhattisgarh, India | +91-6260412013 | anuragsahu812000@gmail.com
linkedin.com/in/anuragkumarsahu1 | github.com/Anurag812000

SUMMARY

GATE-qualified Computer Science Engineer pursuing an M.Tech in Computer Science and Engineering with hands-on experience in **Python programming, software development, machine learning, REST APIs, data processing, and AI engineering**. Strong foundation in **programming fundamentals, data structures, algorithms, object-oriented programming, analytical problem solving, debugging, and testing**. Built practical AI applications using **FastAPI, LangChain, RAG, Qdrant, LLMs, embeddings, and AI agents**.

TECHNICAL SKILLS

Programming: Python, C, C++, SQL, Object-Oriented Programming, Data Structures, Algorithms, Problem Solving, Computational Complexity

AI & Generative AI: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Prompt Engineering, AI Agents, LangChain, Embeddings, Qdrant, Scikit-Learn, PyTorch, TensorFlow

Data & Python Libraries: Pandas, NumPy, Data Processing, Data Preprocessing, Feature Engineering, EDA, Statistical Analysis, Data Analytics, Data Visualization, Large-Scale Dataset Processing

Backend & Software: FastAPI, REST APIs, API Integration, PostgreSQL, MQTT, Git, GitHub, GitHub Actions, Docker, CI/CD, Unit Testing, Debugging

Computer Vision & Tools: OpenCV, Ultralytics, YOLOv8, Label Studio, Object Detection, Instance Segmentation, Jupyter Notebook, Google Colab, VS Code, AWS EC2/Lambda/CloudWatch

EXPERIENCE

Robotic Programmer Intern | Janyu Tech Aug 2026 – Present

- **YOLOv8 Computer Vision & Robotic Automation:** Developed an instance segmentation pipeline using **Python, PyTorch, Ultralytics, YOLOv8, and OpenCV** for identifying and segmenting objects from images and live camera feeds. Created a custom dataset of **198 images and 1,368 polygon-annotated instances** using Label Studio and trained a lightweight YOLOv8-nano model, achieving **0.84 mAP50 detection** and **0.81 mAP50 segmentation**.
- **Varaha IoT – Real-Time Monitoring Dashboard:** Developed an end-to-end telemetry monitoring solution using **MQTT, Node.js, PostgreSQL, React, and TypeScript** for real-time visualization of motor health, RPM, current, cleanliness level, remaining capacity, and tank/device status. Implemented MQTT-based telemetry simulation and verification workflows for real-time data ingestion and backend services.

Machine Learning Intern | DevTown (Remote) Jan 2023 – Apr 2023

- Developed and tested AI-based **potato plant disease detection** using **Python, TensorFlow, and CNNs**, covering data acquisition, image preprocessing, model training, validation, and deployment.
- Developed **FastAPI REST APIs** for deployable ML applications and real-time inference; performed debugging, testing, validation, and performance optimization during development.
- Worked across end-to-end ML workflows and collaborated on development, testing, documentation, and project execution.

PROJECTS

Intelligent Document Q&A and AI Agent | Python, FastAPI, LangChain, RAG, Qdrant, LLM 2026

- Built a Python-based **Retrieval-Augmented Generation (RAG)** application using **FastAPI, LangChain, embeddings, and Qdrant** to ingest documents and generate citation-grounded responses.
- Implemented **document chunking, vector retrieval, prompt construction, and LLM-based answer generation** with source attribution; exposed the pipeline through REST APIs.
- Developed an **AI agent** capable of selecting between document retrieval and utility tools based on the user query.
- Implemented evaluation for **retrieval quality, response correctness, citation correctness, and latency**, with modular components designed for testing and extension.

RESEARCH

Explainable XGBoost-Based Smart Grid Load Forecasting for 6G-Enabled Cyber-Physical Systems

- Developed an explainable **XGBoost** forecasting framework using derivative-based feature engineering, lag variables, rolling statistics, cyclical time encoding, and surge ratio; used a **strict chronological train-test split** with training-only preprocessing to prevent temporal data leakage.
- Achieved **MAE = 1.04, RMSE = 1.39,** R^2 **= 0.836** and applied **Temporal TreeSHAP** over a 168-hour high-stress period to interpret feature contributions during load peaks and abnormal grid conditions.

Lightweight Multivariate Predictive Auto-Scaling for Energy-Efficient Cloud Data Centers

- Designed a lightweight **MAPE-K-based Predictor–Guardian framework** integrating workload prediction and anomalous traffic detection using large-scale NASA HTTP request data.
- Evaluated machine-learning models including Random Forest and Decision Tree, achieving R^2 **= 0.8400** and **F1 = 0.9955**, respectively, for the evaluated tasks.

EDUCATION

International Institute of Information Technology, Naya Raipur 2025 – Present
M.Tech in Computer Science and Engineering **CGPA: 7.8/10**

Bachelor of Technology – Data Science **CGPA: 8.0/10**

ACHIEVEMENTS & LEADERSHIP

GATE CSE Qualified | Placement Coordinator | IIIT Naya Raipur – Coordinated industry interactions, recruitment drives, student engagement, and communication with company representatives and placement teams.

CERTIFICATIONS

AWS Cloud Fundamentals | Python for Data Science | Machine Learning and Deep Learning Certifications